



DW_inc_gray

Gray Incrementer

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History

Description

DW_inc_gray adds input at ci to Gray coded input a and produces the Gray coded output z.

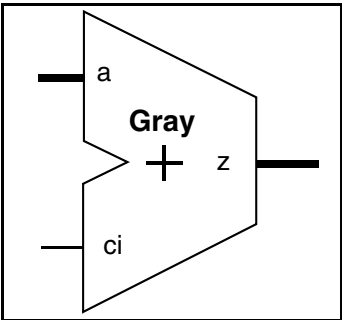


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	width bits	Input	Gray coded input data
ci	1 bit	Input	Carry-in
z	width bits	Output	Gray coded output data

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 1	Input word length

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
rpl	Ripple-carry synthesis model	DesignWare
cla	Carry-lookahead synthesis model	DesignWare

a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the set_implementation command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW01.DW_inc_gray_cfg_sim	Design unit name for VHDL simulation
dw/dw01/src/DW_inc_gray_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_inc_gray.v	Verilog simulation model source code

Table 1-5 Truth Table

a - Gray Input	ci - Carry-in	z - Gray Output
000	0	000
001	0	001
011	0	011
010	0	010
110	0	110
111	0	111
101	0	101
100	0	100
000	1	001
001	1	011
011	1	010
010	1	110
110	1	111
111	1	101
101	1	100
100	1	000

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;  
use IEEE.std_logic_1164.all;  
use DWARE.DW_Foundation_arith.all;  
  
entity DW_inc_gray_func is  
    generic (func_width : positive := 8);  
    port (func_a  : in  std_logic_vector(func_width-1 downto 0);  
          func_ci : in  std_logic;  
          z_func  : out std_logic_vector(func_width-1 downto 0));  
end DW_inc_gray_func;  
  
architecture func of DW_inc_gray_func is  
begin  
    -- function inference of DW_inc_gray  
    z_func <= DWF_inc_gray (func_a, func_ci);  
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_inc_gray_func (func_a, func_ci, z_func);

    parameter func_width = 8;

    input  [func_width-1 : 0] func_a;
    input                        func_ci;
    output [func_width-1 : 0] z_func;

    // pass "width" parameters to the inference functions
    parameter width = func_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW_inc_gray_function.inc"

    // function inference of DW_inc_gray
    assign z_func = DWF_inc_gray (func_a, func_ci);
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_inc_gray_inst is
    generic (inst_width : positive := 8);
    port (inst_a  : in  std_logic_vector(inst_width-1 downto 0);
          inst_ci : in  std_logic;
          z_inst  : out std_logic_vector(inst_width-1 downto 0));
end DW_inc_gray_inst;

architecture inst of DW_inc_gray_inst is
begin
    -- instance of DW_inc_gray
    U1 : DW_inc_gray
        generic map (width => inst_width)
        port map (a  => inst_a,
                  ci => inst_ci,
                  z  => z_inst);
end inst;

-- pragma translate_off
configuration DW_inc_gray_inst_cfg_inst of DW_inc_gray_inst is
    for inst
    end for;
end DW_inc_gray_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_inc_gray_inst (inst_a, inst_ci, z_inst);

    parameter inst_width = 8;

    input  [inst_width-1 : 0] inst_a;
    input                                inst_ci;
    output [inst_width-1 : 0] z_inst;

    // Please add +incdir+$SYNOPTSYS/dw/sim_ver+ to your verilog simulator
    // command line (for simulation).

    // instance of DW_inc_gray
    DW_inc_gray #(inst_width)
        U1 (.a(inst_a),
            .ci(inst_ci),
            .z(z_inst));
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

Copyright Notice and Proprietary Information

© 2024 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.
www.synopsys.com